	$\sigma^{\#1}_{1^+\,\alpha\beta}$	$\sigma^{\#2}_{1^+\,\alpha\beta}$	$\tau^{\#1}_{1^+\,\alpha\beta}$	$\sigma^{\#1}_{1^-\,\alpha}$	$\sigma^{\#2}_{1^-\,\alpha}$	$\tau^{\#1}_{1^-\,\alpha}$	$\tau^{\#2}_{1^-\,\alpha}$
$\sigma^{\#1}_{1^+}\dagger^{\alpha\beta}$	$\frac{(2\beta_1-\beta_2)(1+k^2)}{\mathrm{Det}(1^+)}$	$-\frac{4\beta_1-6\beta_2+(\mathcal{M}_{Pl}^2)}{2\sqrt{2}\,\mathrm{Det}(1^+)}$	$-\frac{i k(4\beta_1-6\beta_2+(\mathcal{M}_{Pl}^2))}{2\sqrt{2}\,\mathrm{Det}(1^+)}$	0	0	0	0
$\sigma^{\#2}_{1^+}\dagger^{\alpha\beta}$	$-\frac{4\beta_1-6\beta_2+(\mathcal{M}_{Pl}^2)}{2\sqrt{2}\,\mathrm{Det}(1^+)}$	$\frac{12\beta_1-10\beta_2+2(\alpha_2-\alpha_3+4\alpha_4-4\alpha_6)k^2+(\mathcal{M}_{Pl}^2)}{4(1+k^2)\,\mathrm{Det}(1^+)}$	$\frac{i k(12\beta_1-10\beta_2+2(\alpha_2-\alpha_3+4\alpha_4-4\alpha_6)k^2+(\mathcal{M}_{Pl}^2))}{4(1+k^2)\,\mathrm{Det}(1^+)}$	0	0	0	0
$\tau^{\#1}_{1^+}\dagger^{\alpha\beta}$	$\frac{i k(4\beta_1-6\beta_2+(\mathcal{M}_{Pl}^2))}{2\sqrt{2}\,\mathrm{Det}(1^+)}$	$-\frac{i k(12\beta_1-10\beta_2+2(\alpha_2-\alpha_3+4\alpha_4-4\alpha_6)k^2+(\mathcal{M}_{Pl}^2))}{4(1+k^2)\,\mathrm{Det}(1^+)}$	$\frac{k^2(12\beta_1-10\beta_2+2(\alpha_2-\alpha_3+4\alpha_4-4\alpha_6)k^2+(\mathcal{M}_{Pl}^2))}{4(1+k^2)\,\mathrm{Det}(1^+)}$	0	0	0	0
$\sigma^{\#1}_{1^-}\dagger^{\alpha}$	0	0	0	$\frac{(1+2k^2)(72(2\beta_1+\beta_2+\beta_3)+k^2\xi)}{144\,\mathrm{Det}(1^-)}$	$\frac{72\beta_3+36(\mathcal{M}_{Pl}^2)+k^2(6\chi+\xi)}{72\sqrt{2}\,\mathrm{Det}(1^-)}$	0	$\frac{i k(72\beta_3+36(\mathcal{M}_{Pl}^2)+k^2(6\chi+\xi))}{72\,\mathrm{Det}(1^-)}$
$\sigma^{\#2}_{1^-}\dagger^{\alpha}$	0	0	0	$\frac{72\beta_3+36(\mathcal{M}_{Pl}^2)+k^2(6\chi+\xi)}{72\sqrt{2}\,\mathrm{Det}(1^-)}$	$\frac{72\beta_1+36\beta_2+72\beta_3+18(\mathcal{M}_{Pl}^2)+k^2(12(6\alpha_2+12\alpha_4+3\alpha_5+\chi)+\xi)}{72(1+2k^2)\,\mathrm{Det}(1^-)}$	0	$\frac{i k(72\beta_1+36\beta_2+72\beta_3+18(\mathcal{M}_{Pl}^2)+k^2(12(6\alpha_2+12\alpha_4+3\alpha_5+\chi)+\xi))}{36\sqrt{2}(1+2k^2)\,\mathrm{Det}(1^-)}$
$\tau^{\#1}_{1^-}\dagger^{\alpha}$	0	0	0	0	0	0	0
$\tau^{\#2}_{1^-}\dagger^{\alpha}$	0	0	0	$-\frac{i k(72\beta_3+36(\mathcal{M}_{Pl}^2)+k^2(6\chi+\xi))}{72\,\mathrm{Det}(1^-)}$	$-\frac{i k(72\beta_1+36\beta_2+72\beta_3+18(\mathcal{M}_{Pl}^2)+k^2(12(6\alpha_2+12\alpha_4+3\alpha_5+\chi)+\xi))}{36\sqrt{2}(1+2k^2)\,\mathrm{Det}(1^-)}$	0	$\frac{6k^2(6(2\beta_1+\beta_2+2\beta_3)+2(6\alpha_2+12\alpha_4+3\alpha_5+\chi)k^2+3(\mathcal{M}_{Pl}^2))+k^4\xi}{(36+72k^2)\,\mathrm{Det}(1^-)}$

	$\mathcal{A}^{\#1}_{1^+\,\alpha\beta}$	$\mathcal{A}^{\#2}_{1^+\,\alpha\beta}$	$f^{\#1}_{1^+\,\alpha\beta}$	$\mathcal{A}^{\#1}_{1^-\,\alpha}$	$\mathcal{A}^{\#2}_{1^-\,\alpha}$	$f^{\#1}_{1^-\,\alpha}$	$f^{\#2}_{1^-\,\alpha}$
$\mathcal{A}^{\#1}_{1^+}\dagger^{\alpha\beta}$	$\frac{1}{4}(12\beta_1-10\beta_2+2(\alpha_2-\alpha_3+4\alpha_4-4\alpha_6)k^2+(\mathcal{M}_{Pl}^2))$	$\frac{4\beta_1-6\beta_2+(\mathcal{M}_{Pl}^2)}{2\sqrt{2}}$	$\frac{i k(4\beta_1-6\beta_2+(\mathcal{M}_{Pl}^2))}{2\sqrt{2}}$	0	0	0	0
$\mathcal{A}^{\#2}_{1^+}\dagger^{\alpha\beta}$	$\frac{4\beta_1-6\beta_2+(\mathcal{M}_{Pl}^2)}{2\sqrt{2}}$	$2\beta_1-\beta_2$	$i(2\beta_1-\beta_2)k$	0	0	0	0
$f^{\#1}_{1^+}\dagger^{\alpha\beta}$	$-\frac{i k(4\beta_1-6\beta_2+(\mathcal{M}_{Pl}^2))}{2\sqrt{2}}$	$-i(2\beta_1-\beta_2)k$	$(2\beta_1-\beta_2)k^2$	0	0	0	0
$\mathcal{A}^{\#1}_{1^-}\dagger^{\alpha}$	0	0	0	$\beta_1+\frac{\beta_2}{2}+\beta_3+\frac{(\mathcal{M}_{Pl}^2)}{4}+\frac{1}{72}k^2(12(6\alpha_2+12\alpha_4+3\alpha_5+\chi)+\xi)$	$-\frac{72\beta_3+36(\mathcal{M}_{Pl}^2)+k^2(6\chi+\xi)}{72\sqrt{2}}$	0	$-\frac{1}{72}i k(72\beta_3+36(\mathcal{M}_{Pl}^2)+k^2(6\chi+\xi))$
$\mathcal{A}^{\#2}_{1^-}\dagger^{\alpha}$	0	0	0	$-\frac{72\beta_3+36(\mathcal{M}_{Pl}^2)+k^2(6\chi+\xi)}{72\sqrt{2}}$	$\beta_1+\frac{\beta_2+\beta_3}{2}+\frac{k^2\xi}{144}$	0	$\frac{i k(72(2\beta_1+\beta_2+\beta_3)+k^2\xi)}{72\sqrt{2}}$
$f^{\#1}_{1^-}\dagger^{\alpha}$	0	0	0	0	0	0	0
$f^{\#2}_{1^-}\dagger^{\alpha}$	0	0	0	$\frac{1}{72}i k(72\beta_3+36(\mathcal{M}_{Pl}^2)+k^2(6\chi+\xi))$	$-\frac{i k(72(2\beta_1+\beta_2+\beta_3)+k^2\xi)}{72\sqrt{2}}$	0	$(2\beta_1+\beta_2+\beta_3)k^2+\frac{k^4\xi}{72}$

Source constraints

SO(3) irreps	#
$\tau^{\#2}_{0^+}==0$	1
$\tau^{\#2\alpha}_{1^-}+2 i k\,\sigma^{\#2\alpha}_{1^-}==0$	3
$\tau^{\#1\alpha}_{1^-}==0$	3
$\tau^{\#1\alpha\beta}_{1^+}+i k\,\sigma^{\#2\alpha\beta}_{1^+}==0$	3
Total #:	10

	$\mathcal{A}^{\#1}_{0^+}$	$f^{\#1}_{0^+}$	$f^{\#2}_{0^+}$	$\mathcal{A}^{\#1}_{0^-\,\alpha\beta\chi}$
$\mathcal{A}^{\#1}_{0^+}\dagger$	$\frac{1}{2}(2\beta_1+\beta_2+3\beta_3+2(6\alpha_1+2\alpha_2+2\alpha_3+2\alpha_4+\alpha_5+2\alpha_6)k^2+(\mathcal{M}_{Pl}^2))$	$-\frac{i k(2\beta_1+\beta_2+3\beta_3+(\mathcal{M}_{Pl}^2))}{\sqrt{2}}$	0	0
$f^{\#1}_{0^+}\dagger$	$\frac{i k(2\beta_1+\beta_2+3\beta_3+(\mathcal{M}_{Pl}^2))}{\sqrt{2}}$	$(2\beta_1+\beta_2+3\beta_3)k^2$	0	0
$f^{\#2}_{0^+}\dagger$	0	0	0	0
$\mathcal{A}^{\#1}_{0^-}\dagger^{\alpha\beta\chi}$	0	0	0	$\frac{1}{2}(8\beta_1-8\beta_2+4\alpha_4k^2-2\alpha_5k^2+(\mathcal{M}_{Pl}^2))$

$$\mathrm{Det}(2^-)\to\frac{1}{4}(4\beta_1+2\beta_2+2(4\alpha_4+\alpha_5)k^2-(\mathcal{M}_{Pl}^2))$$

$$\mathrm{Det}(0^-)\to\frac{1}{2}(8\beta_1-8\beta_2+4\alpha_4k^2-2\alpha_5k^2+(\mathcal{M}_{Pl}^2))$$

	$\mathcal{A}^{\#1}_{2^+\,\alpha\beta}$	$f^{\#1}_{2^+\,\alpha\beta}$	$\mathcal{A}^{\#1}_{2^-\,\alpha\beta\chi}$
$\mathcal{A}^{\#1}_{2^+}\dagger^{\alpha\beta}$	$\beta_1+\frac{1}{2}(\beta_2+(\alpha_2+\alpha_3+4\alpha_4+2\alpha_5+4\alpha_6)k^2)-\frac{(\mathcal{M}_{Pl}^2)}{4}$	$-\frac{i k(4\beta_1+2\beta_2-(\mathcal{M}_{Pl}^2))}{2\sqrt{2}}$	0
$f^{\#1}_{2^+}\dagger^{\alpha\beta}$	$\frac{i k(4\beta_1+2\beta_2-(\mathcal{M}_{Pl}^2))}{2\sqrt{2}}$	$(2\beta_1+\beta_2)k^2$	0
$\mathcal{A}^{\#1}_{2^-}\dagger^{\alpha\beta\chi}$	0	0	$\frac{1}{4}(4\beta_1+2\beta_2+2(4\alpha_4+\alpha_5)k^2-(\mathcal{M}_{Pl}^2))$

$$\mathrm{Det}(0^+)\to\frac{1}{2}k^2(2(6\alpha_1+2\alpha_2+2\alpha_3+2\alpha_4+\alpha_5+2\alpha_6)(2\beta_1+\beta_2+3\beta_3)k^2-(2\beta_1+\beta_2+3\beta_3)(\mathcal{M}_{Pl}^2)-(\mathcal{M}_{Pl}^2)^2)$$

$$\mathrm{Det}(2^+)\to\frac{1}{8}k^2(4(\alpha_2+\alpha_3+4\alpha_4+2\alpha_5+4\alpha_6)(2\beta_1+\beta_2)k^2+2(2\beta_1+\beta_2)(\mathcal{M}_{Pl}^2)-(\mathcal{M}_{Pl}^2)^2)$$

$$\mathrm{Det}(1^-)\to\frac{1}{576}(1+2k^2)(2(288\beta_1^2+72\beta_2^2+72(2\alpha_2+4\alpha_4+\alpha_5)\beta_3k^2-\chi^2k^4+48\beta_1(6\beta_2+9\beta_3+(6\alpha_2+12\alpha_4+3\alpha_5+\chi)k^2)+72\beta_1(\mathcal{M}_{Pl}^2)-12(9\beta_3+\chi k^2)(\mathcal{M}_{Pl}^2)-36(\mathcal{M}_{Pl}^2)^2+12\beta_2(18\beta_3+2(6\alpha_2+12\alpha_4+3\alpha_5+\chi)k^2+3(\mathcal{M}_{Pl}^2)))+k^2(12\beta_1+6\beta_2+2(2\alpha_2+4\alpha_4+\alpha_5)k^2-3(\mathcal{M}_{Pl}^2))\xi)$$

	$\sigma^{\#1}_{0^+}$	$\tau^{\#1}_{0^+}$	$\tau^{\#2}_{0^+}$	$\sigma^{\#1}_{0^-\,\alpha\beta\chi}$
$\sigma^{\#1}_{0^+}\dagger$	$\frac{(2\beta_1+\beta_2+3\beta_3)k^2}{\mathrm{Det}(0^+)}$	$\frac{i k(2\beta_1+\beta_2+3\beta_3+(\mathcal{M}_{Pl}^2))}{\sqrt{2}\,\mathrm{Det}(0^+)}$	0	0
$\tau^{\#1}_{0^+}\dagger$	$-\frac{i k(2\beta_1+\beta_2+3\beta_3+(\mathcal{M}_{Pl}^2))}{\sqrt{2}\,\mathrm{Det}(0^+)}$	$\frac{2\beta_1+\beta_2+3\beta_3+2(6\alpha_1+2\alpha_2+2\alpha_3+2\alpha_4+\alpha_5+2\alpha_6)k^2+(\mathcal{M}_{Pl}^2)}{2\,\mathrm{Det}(0^+)}$	0	0
$\tau^{\#2}_{0^+}\dagger$	0	0	0	0
$\sigma^{\#1}_{0^-}\dagger^{\alpha\beta\chi}$	0	0	0	$\frac{1}{\mathrm{Det}(0^-)}$

	$\sigma^{\#1}_{2^+\,\alpha\beta}$	$\tau^{\#1}_{2^+\,\alpha\beta}$	$\sigma^{\#1}_{2^-\,\alpha\beta\chi}$
$\sigma^{\#1}_{2^+}\dagger^{\alpha\beta}$	$\frac{(2\beta_1+\beta_2)k^2}{\mathrm{Det}(2^+)}$	$\frac{i k(4\beta_1+2\beta_2-(\mathcal{M}_{Pl}^2))}{2\sqrt{2}\,\mathrm{Det}(2^+)}$	0
$\tau^{\#1}_{2^+}\dagger^{\alpha\beta}$	$-\frac{i k(4\beta_1+2\beta_2-(\mathcal{M}_{Pl}^2))}{2\sqrt{2}\,\mathrm{Det}(2^+)}$	$\frac{\beta_1+\frac{1}{2}(\beta_2+(\alpha_2+\alpha_3+4\alpha_4+2\alpha_5+4\alpha_6)k^2)-\frac{(\mathcal{M}_{Pl}^2)}{4}}{\mathrm{Det}(2^+)}$	0
$\sigma^{\#1}_{2^-}\dagger^{\alpha\beta\chi}$	0	0	$\frac{1}{\mathrm{Det}(2^-)}$

Lagrangian density

$$2\beta_1\mathcal{A}_{\imath\theta\kappa}\mathcal{A}^{\imath\theta\kappa}-\beta_2\mathcal{A}_{\imath\theta\kappa}\mathcal{A}^{\imath\theta\kappa}-2\beta_1\mathcal{A}_{\imath\kappa\theta}\mathcal{A}^{\imath\theta\kappa}+3\beta_2\mathcal{A}_{\imath\kappa\theta}\mathcal{A}^{\imath\theta\kappa}-\frac{1}{2}(\mathcal{M}_{Pl}^2)\mathcal{A}_{\imath\kappa\theta}\mathcal{A}^{\imath\theta\kappa}-\beta_3\mathcal{A}^{\imath\theta}{}_{\imath}\mathcal{A}_{\theta}{}^{\kappa}{}_{\kappa}-\frac{1}{2}(\mathcal{M}_{Pl}^2)\mathcal{A}^{\imath\theta}{}_{\imath}\mathcal{A}_{\theta}{}^{\kappa}{}_{\kappa}-$$
$$\alpha_2\partial_\imath\mathcal{A}^{\kappa\lambda}{}_{\kappa}\partial^\imath\mathcal{A}_{\lambda}{}^{\zeta}{}_{\zeta}+\beta_3\mathcal{A}_{\imath}{}^{\kappa}{}_{\kappa}\partial_\theta f^{\imath\theta}-\beta_3\mathcal{A}^{\kappa\lambda}{}_{\kappa}\partial_\theta f_{\lambda}{}^{\theta}-(\mathcal{M}_{Pl}^2)f^{\imath\theta}\partial_\theta\mathcal{A}_{\imath}{}^{\kappa}{}_{\kappa}+$$
$$(\mathcal{M}_{Pl}^2)\partial_\theta\mathcal{A}^{\imath\theta}{}_{\imath}-\alpha_5\partial^\alpha\mathcal{A}^{\theta\zeta}{}_{\kappa}\partial_\theta\mathcal{A}_{\zeta\alpha}{}^{\kappa}-\beta_3\mathcal{A}_{\theta}{}^{\kappa}{}_{\kappa}\partial^\theta f^{\imath}{}_{\imath}+\beta_3\mathcal{A}^{\kappa}{}_{\theta\kappa}\partial^\theta f^{\imath}{}_{\imath}+$$
$$\beta_3\partial_\theta f^{\kappa}{}_{\kappa}\partial^\theta f^{\imath}{}_{\imath}+(\mathcal{M}_{Pl}^2)f^{\imath\theta}\partial_\kappa\mathcal{A}_{\imath}{}^{\kappa}{}_{\theta}-(\mathcal{M}_{Pl}^2)f^{\imath}{}_{\imath}\partial_\kappa\mathcal{A}^{\theta\kappa}{}_{\theta}-\alpha_3\partial_\theta\mathcal{A}_{\lambda}{}^{\zeta}{}_{\zeta}\partial_\kappa\mathcal{A}^{\theta\kappa\lambda}+$$
$$\alpha_5\partial_\theta\mathcal{A}_{\zeta\alpha}{}^{\kappa}\partial_\kappa\mathcal{A}^{\theta\zeta\alpha}+2\alpha_2\partial_\theta\mathcal{A}_{\lambda}{}^{\zeta}{}_{\zeta}\partial_\kappa\mathcal{A}^{\kappa\lambda\theta}-2\alpha_4\partial_\theta\mathcal{A}_{\zeta\alpha}{}^{\kappa}\partial_\kappa\mathcal{A}^{\zeta\alpha\theta}+$$
$$2\beta_1\mathcal{A}_{\imath\theta\kappa}\partial^\kappa f^{\imath\theta}-\beta_2\mathcal{A}_{\imath\theta\kappa}\partial^\kappa f^{\imath\theta}+2\beta_2\mathcal{A}_{\imath\kappa}{}^{\lambda}\partial^\kappa f^{\imath}{}_{\lambda}+2\beta_1\mathcal{A}_{\theta\kappa}{}^{\lambda}\partial^\kappa f_{\lambda}{}^{\theta}-$$
$$\beta_2\mathcal{A}_{\theta\kappa}{}^{\lambda}\partial^\kappa f_{\lambda}{}^{\theta}-2\beta_1\mathcal{A}_{\theta}{}^{\lambda}{}_{\kappa}\partial^\kappa f_{\lambda}{}^{\theta}+\beta_2\mathcal{A}_{\theta}{}^{\lambda}{}_{\kappa}\partial^\kappa f_{\lambda}{}^{\theta}+2\beta_2\mathcal{A}^{\lambda}{}_{\kappa\theta}\partial^\kappa f_{\lambda}{}^{\theta}+$$
$$\beta_2\partial_\kappa f_{\theta}{}^{\lambda}\partial^\kappa f_{\lambda}{}^{\theta}+2\beta_1\partial_\kappa f^{\lambda}{}_{\theta}\partial^\kappa f_{\lambda}{}^{\theta}+2\beta_1\mathcal{A}_{\theta\kappa}{}^{\zeta}\partial^\kappa f_{\zeta}{}^{\theta}-$$
$$\beta_2\mathcal{A}_{\theta\kappa}{}^{\zeta}\partial^\kappa f_{\zeta}{}^{\theta}+\frac{1}{6}\chi\partial_\theta\mathcal{A}_{\kappa}{}^{\lambda}{}_{\lambda}\partial^\kappa\mathcal{A}^{\imath\theta}{}_{\imath}+\frac{1}{72}\xi\partial_\theta\mathcal{A}_{\kappa}{}^{\lambda}{}_{\lambda}\partial^\kappa\mathcal{A}^{\imath\theta}{}_{\imath}-$$
$$\frac{1}{6}\chi\partial_\kappa\mathcal{A}_{\theta}{}^{\lambda}{}_{\lambda}\partial^\kappa\mathcal{A}^{\imath\theta}{}_{\imath}-\frac{1}{72}\xi\partial_\kappa\mathcal{A}_{\theta}{}^{\lambda}{}_{\lambda}\partial^\kappa\mathcal{A}^{\imath\theta}{}_{\imath}-\alpha_5\partial_\kappa\mathcal{A}^{\theta\zeta\alpha}\partial^\kappa\mathcal{A}_{\zeta\alpha\theta}+$$
$$2\alpha_4\partial_\kappa\mathcal{A}^{\zeta\alpha\theta}\partial^\kappa\mathcal{A}_{\zeta\alpha\theta}-\frac{1}{72}\xi\partial_\imath\mathcal{A}_{\kappa}{}^{\lambda}{}_{\lambda}\partial^\kappa\partial_\theta f^{\imath\theta}-\frac{1}{6}\chi\partial_\imath\mathcal{A}_{\kappa}{}^{\zeta}{}_{\zeta}\partial^\kappa\partial_\theta f^{\imath\theta}-$$
$$\frac{1}{72}\xi\partial_\imath\mathcal{A}_{\kappa}{}^{\zeta}{}_{\zeta}\partial^\kappa\partial_\theta f^{\imath\theta}+\frac{1}{72}\xi\partial_\kappa\mathcal{A}_{\imath}{}^{\lambda}{}_{\lambda}\partial^\kappa\partial_\theta f^{\imath\theta}+\frac{1}{6}\chi\partial_\kappa\mathcal{A}_{\imath}{}^{\zeta}{}_{\zeta}\partial^\kappa\partial_\theta f^{\imath\theta}+$$
$$\frac{1}{72}\xi\partial_\kappa\mathcal{A}_{\imath}{}^{\zeta}{}_{\zeta}\partial^\kappa\partial_\theta f^{\imath\theta}+\frac{1}{6}\chi\partial_\theta\mathcal{A}_{\kappa}{}^{\zeta}{}_{\zeta}\partial^\kappa\partial^\theta f^{\imath}{}_{\imath}+\frac{1}{72}\xi\partial_\theta\mathcal{A}_{\kappa}{}^{\zeta}{}_{\zeta}\partial^\kappa\partial^\theta f^{\imath}{}_{\imath}-$$
$$\frac{1}{6}\chi\partial_\kappa\mathcal{A}_{\theta}{}^{\zeta}{}_{\zeta}\partial^\kappa\partial^\theta f^{\imath}{}_{\imath}-\frac{1}{72}\xi\partial_\kappa\mathcal{A}_{\theta}{}^{\zeta}{}_{\zeta}\partial^\kappa\partial^\theta f^{\imath}{}_{\imath}-2\beta_3\partial^\theta f^{\imath}{}_{\imath}\partial_\lambda f_{\theta}{}^{\lambda}+$$
$$\beta_3\partial_\theta f_{\kappa}{}^{\theta}\partial_\lambda f^{\kappa\lambda}+\frac{1}{6}\chi\partial^\kappa\mathcal{A}^{\imath\theta}{}_{\imath}\partial_\lambda\mathcal{A}_{\theta}{}^{\lambda}{}_{\kappa}-\frac{1}{6}\chi\partial^\kappa\mathcal{A}^{\imath\theta}{}_{\imath}\partial_\lambda\mathcal{A}_{\kappa}{}^{\lambda}{}_{\theta}+$$
$$4\alpha_1\partial_\theta\mathcal{A}^{\imath\theta}{}_{\imath}\partial_\lambda\mathcal{A}^{\kappa\lambda}{}_{\kappa}+4\alpha_6\partial^\alpha\mathcal{A}_{\imath}{}^{\lambda\zeta}\partial_\lambda\mathcal{A}_{\zeta\alpha}{}^{\imath}+\alpha_5\partial^\alpha\mathcal{A}_{\imath}{}^{\zeta\lambda}\partial_\lambda\mathcal{A}_{\zeta\alpha}{}^{\imath}+$$
$$\alpha_3\partial_\theta\mathcal{A}_{\lambda}{}^{\zeta}{}_{\zeta}\partial^\lambda\mathcal{A}^{\theta\kappa}{}_{\kappa}-\frac{1}{6}\chi\partial^\kappa\partial_\theta f^{\imath\theta}\partial_\zeta\mathcal{A}_{\imath}{}^{\zeta}{}_{\kappa}+\frac{1}{6}\chi\partial^\kappa\partial^\theta f^{\imath}{}_{\imath}\partial_\zeta\mathcal{A}_{\theta}{}^{\zeta}{}_{\kappa}+$$
$$\frac{1}{6}\chi\partial^\kappa\partial_\theta f^{\imath\theta}\partial_\zeta\mathcal{A}_{\kappa}{}^{\zeta}{}_{\imath}-\frac{1}{6}\chi\partial^\kappa\partial^\theta f^{\imath}{}_{\imath}\partial_\zeta\mathcal{A}_{\kappa}{}^{\zeta}{}_{\theta}+\alpha_3\partial_\kappa\mathcal{A}^{\theta\kappa\lambda}\partial_\zeta\mathcal{A}_{\lambda}{}^{\zeta}{}_{\theta}-$$
$$\alpha_2\partial_\kappa\mathcal{A}^{\kappa\lambda\theta}\partial_\zeta\mathcal{A}_{\lambda}{}^{\zeta}{}_{\theta}-\alpha_3\partial^\lambda\mathcal{A}^{\theta\kappa}{}_{\kappa}\partial_\zeta\mathcal{A}_{\lambda}{}^{\zeta}{}_{\theta}-\frac{1}{72}\xi\partial^\kappa\partial_\theta f^{\imath\theta}\partial_\zeta\partial_\imath f_{\kappa}{}^{\zeta}+$$
$$\frac{1}{72}\xi\partial^\kappa\partial^\theta f^{\imath}{}_{\imath}\partial_\zeta\partial_\theta f_{\kappa}{}^{\zeta}+\frac{1}{72}\xi\partial^\kappa\partial_\theta f^{\imath\theta}\partial_\zeta\partial_\kappa f_{\imath}{}^{\zeta}-\frac{1}{72}\xi\partial^\kappa\partial^\theta f^{\imath}{}_{\imath}\partial_\zeta\partial_\kappa f_{\theta}{}^{\zeta}-$$
$$\beta_2\partial^\kappa f_{\zeta}{}^{\theta}\partial^\zeta f_{\theta\kappa}-2\beta_1\partial^\kappa f_{\zeta}{}^{\theta}\partial^\zeta f_{\kappa\theta}+\beta_2\partial^\kappa f_{\lambda\zeta}\partial^\zeta f^{\lambda}{}_{\kappa}-\beta_2\partial^\kappa f_{\zeta\lambda}\partial^\zeta f^{\lambda}{}_{\kappa}$$

Added source term(s): $f^{\alpha\beta}\,\tau_{\alpha\beta}+\mathcal{A}^{\alpha\beta\chi}\,\sigma_{\alpha\beta\chi}$

$$\mathrm{Det}(1^+)\to\frac{1}{8}(1+k^2)(16(\beta_1-\beta_2)(2\beta_1+\beta_2)+4(\alpha_2-\alpha_3+4\alpha_4-4\alpha_6)(2\beta_1-\beta_2)k^2-4\beta_1(\mathcal{M}_{Pl}^2)+10\beta_2(\mathcal{M}_{Pl}^2)-(\mathcal{M}_{Pl}^2)^2)$$